

**Determining the neural circuits supporting reading in children with left vs right
amblyopia
("lazy eye") using EEG**

Introduction

Amblyopia, commonly referred to as “lazy eye,” is a neurodevelopmental visual disorder characterized by reduced visual acuity that cannot be explained by structural abnormalities of the eye alone. The condition typically develops during early childhood as a result of abnormal visual experience, including strabismus, refractive imbalance, or visual deprivation. Beyond reduced acuity, amblyopia has been associated with deficits in binocular vision, oculomotor control, visual attention, and higher-order visual processing.

Recent evidence suggests that amblyopia may also influence reading fluency and audiovisual integration processes. Efficient reading relies on accurate and temporally synchronized visual and auditory processing, requiring coordination between visual perception, eye movements, attention, executive functions, and language-related neural networks. Children with amblyopia often demonstrate slower reading speed and altered visual scanning patterns, potentially reflecting differences in the neural mechanisms supporting reading acquisition and fluency.

In particular, differences between left-eye and right-eye amblyopia may be relevant for understanding reading-related neural processing. Since reading depends on lateralized visual and language networks, asymmetric visual input may influence how information is integrated across hemispheres during reading tasks. Previous studies have suggested that amblyopia may affect functional connectivity, attentional engagement, and audiovisual synchronization, highlighting the importance of investigating neural activity alongside behavioral performance.

The current project focuses on the development of an integrated experimental and engineering platform designed to synchronize behavioral, physiological, and neurophysiological measurements during reading tasks. The system combines EEG, eye-tracking, and heart-rate acquisition with controlled visual and auditory experimental paradigms in order to investigate reading fluency, visual processing, and audiovisual integration in children with amblyopia.

This project book serves both as research documentation and as a technical system-design reference. In addition to describing the scientific rationale and experimental structure, it documents the architecture of the experimental interface, synchronization logic, signal acquisition pipelines, preprocessing methods, and analytical components used throughout the project.

Experimental Procedure

The experimental session begins with the participant seated in front of a computer screen. The child is connected to a 64-channel EEG system, a Corsano wrist device for heart-rate and HRV recording, and a Tobii X3-120 eye-tracking system. After EEG and physiological sensors are connected, the eye-tracker calibration procedure is performed to ensure accurate gaze recording throughout the reading tasks.

1. Oral Reading - Erased Text

participants read the text aloud while letters gradually disappear from the screen over time. This condition combines active vocal reading with dynamic visual pacing.

2. Silent Reading - Erased Text

participants read the text silently while letters gradually disappear from the screen. No auditory pacing cues are presented during this task.

3. Silent Reading - Erased Text with Metronome

participants read silently while letters gradually disappear from the screen and rhythmic metronome clicks are simultaneously presented in the background. This condition is designed to examine the interaction between visual pacing and external auditory timing cues.

4. Silent Reading - Static Text with Metronome

participants read silently while the full text remains continuously visible on the screen. Rhythmic metronome clicks are presented throughout the task.

5. Silent Reading - Static Text

participants read silently while the full text remains continuously visible on the screen.

6. Resting state

In addition to the reading tasks, the experiment includes a two-minute resting-state baseline condition, during which the child looks at the screen without performing an active task. This baseline is used to characterize resting neural and physiological activity and to provide a comparison point for task-related EEG, eye-tracking, and heart-rate measures.

After each reading task, the child answers multiple-choice comprehension questions about the text. These questions are used to verify reading engagement and to provide behavioral measures of comprehension for each experimental condition.

Experimental System Structure

The experimental paradigm is controlled using multiple synchronized acquisition systems distributed across separate computers.

The reading tasks are presented using E-Prime, which runs on the participant-facing laptop. This computer is also used for the eye-tracking system, including both the Tobii X3-120 device and the Tobii Pro Lab software used for gaze recording and data acquisition. Since the eye-tracking system must remain directly synchronized with stimulus presentation, E-Prime and Tobii Pro Lab are operated on the same computer.

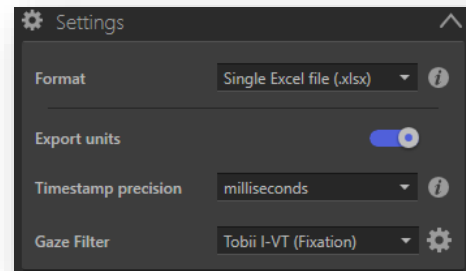
EEG acquisition is performed on a separate recording computer connected to the 64-channel EEG system. The reading tasks used in the current experiment do not require direct event-trigger synchronization between E-Prime and the EEG acquisition computer, allowing the EEG recording system to operate independently during the tasks.

Physiological heart-rate and HRV measurements are collected using the Corsano wearable device. Data from the Corsano device are recorded directly through the Corsano online platform and therefore do not depend on a dedicated local acquisition computer.

This distributed system architecture enables simultaneous acquisition of behavioral, gaze, neural, and physiological data during the experimental session while maintaining separation between the different recording systems.

Eye-Tracking Database: Relevant Fields

Export a file from the software "Tobii pro lab" using the following format:



1. Timing / Time Fields

- Recording timestamp [ms]
- Computer timestamp [ms]
- Eyetracker timestamp [μ s]
- Recording duration [ms]

2. Participant / Recording Metadata

- Participant name
- Recording name
- Recording date
- Recording start time
- Project name
- Timeline name
- Recording software version

3. Calibration / Validation Quality

- Average calibration accuracy [$^{\circ}$]
- Average calibration precision SD [$^{\circ}$]
- Average calibration precision RMS [$^{\circ}$]
- Average validation accuracy [$^{\circ}$]
- Average validation precision SD [$^{\circ}$]
- Average validation precision RMS [$^{\circ}$]

4. Gaze Coordinates

- Gaze point X [DACS px]
- Gaze point Y [DACS px]
- Gaze point left X [DACS px]
- Gaze point left Y [DACS px]
- Gaze point right X [DACS px]
- Gaze point right Y [DACS px]

5. Validity

- Validity left
- Validity right

6. Pupil Diameter

- Pupil diameter left [mm]
- Pupil diameter right [mm]

7. Eye Position

- Eye position left X [DACS mm]
- Eye position left Y [DACS mm]

- Eye position left Z [DACS mm]
- Eye position right X [DACS mm]
- Eye position right Y [DACS mm]
- Eye position right Z [DACS mm]

8. Stimulus / Media Information

- Presented Stimulus name
- Presented Media name
- Presented Media width [px]
- Presented Media height [px]
- Presented Media position X [DACS px]
- Presented Media position Y [DACS px]

9. Eye Movement Events

- Eye movement type
- Gaze event duration [ms]
- Eye movement type index

10. Fixation Fields

- Fixation point X [DACS px]
- Fixation point Y [DACS px]
- Fixation point X [MCS norm]
- Fixation point Y [MCS norm]

11. Event Fields

- Event
- Event value

Eye-Tracking Analysis Fields

Eye movement type

Use: Primary categorical field for gaze-event analysis.

Values: Fixation, Saccade, Unclassified, EyesNotFound.

Analysis use: classify reading behavior into stable visual processing, gaze transitions, tracking loss, and non-classified segments.

Processing needed: count events per task and calculate percentage of total task time per event type.

Validation: exclude or flag segments with high EyesNotFound or high Unclassified.

Gaze event duration [ms]

Use: Primary duration field for each gaze event.

Analysis use: calculate mean fixation duration, total fixation time, mean saccade duration, total tracking-loss duration.

Processing needed: group by Eye movement type, Eye movement type index, participant, and task.

Eye movement type index

Use: Event identifier.

Analysis use: reconstruct event sequence and avoid counting the same fixation/saccade multiple times across rows.

Processing needed: aggregate rows by event index before calculating event-level metrics.

Limitations: should not be treated as a biological measure; it is only an indexing field.

Validity left

Use: left-eye data validity indicator.

Analysis use: quality control and exclusion/flagging of unreliable samples.

Processing needed: calculate percentage of valid samples per task.

Limitations: validity is eye-specific and should be interpreted together with Validity right.

Validity right

Use: right-eye data validity indicator.

Analysis use: quality control and tracking reliability.

Processing needed: calculate valid/invalid sample ratio.

Limitations: important in amblyopia because one eye may be less reliable depending on calibration/tracking.

Pupil diameter left [mm]

Use: left pupil size.

Analysis use: exploratory marker of arousal/cognitive load.

Processing needed: remove invalid samples/blinks and compare task averages.

Limitations: highly sensitive to lighting, tracking loss, and eye validity.

Pupil diameter right [mm]

Use: right pupil size.

Analysis use: exploratory physiological/attention marker.

Processing needed: same cleaning as left pupil.

Limitations: highly sensitive to lighting, tracking loss, and eye validity.

Derived Feature: Regression Detection

Tobii does not directly classify regressions. Regressions must be computed from the eye-tracking event sequence.

For Hebrew reading, the expected reading direction within a line is right-to-left. Therefore, a regression can be operationally defined as a saccade in which gaze moves opposite to the reading direction.

Required Fields

Eye movement type

Eye movement type index

Fixation point X [DACS px]

Fixation point Y [DACS px]

Gaze event duration [ms]

Basic Algorithm

1. Keep only events classified as Fixation and Saccade.
2. Reconstruct the event sequence using Eye movement type index.
3. For each saccade, identify the fixation before and after the saccade.
4. Calculate horizontal displacement:

$\text{delta_x} = \text{next_fixation_x} - \text{previous_fixation_x}$

5. For Hebrew text:

$\text{delta_x} < 0 \rightarrow \text{forward reading movement}$

$\text{delta_x} > 0 \rightarrow \text{regression}$

Important Limitation

This rule should be applied mainly **within the same text line**.

Line transitions can also involve large X changes and may be falsely classified as regressions unless line position is considered using Fixation point Y.

So the better version is:

if same_line == True and delta_x > threshold:

classify as regression

where threshold prevents tiny noisy movements from being counted as regressions.

Feature 1 - Fixation Duration

Classification: Visual + Cognitive

Definition

Duration of a fixation event during reading.

Computed from:

Eye movement type = Fixation

Gaze event duration [ms]

Interpretation

Visual Component

Longer fixations may indicate:

- visual decoding difficulty
 - unstable visual processing
 - increased visual effort
-

Cognitive Component

Longer fixations may also reflect:

- increased cognitive load
 - lexical/semantic processing difficulty
 - attentional slowing
 - working-memory demands
-

Task-Level Analysis

mean_fixation_duration

fixation_time_percentage

fixation_duration_variability

Usage

Compare fixation behavior across:

- amblyopia vs controls
 - left vs right amblyopia
 - task vs task
-

Event-Level Analysis

Definition

Fixations that significantly deviate from the participant/task baseline.

Example:

long_fixation = fixation_duration > participant_mean

Usage

Event-centered multimodal analysis:

- EEG activity during long fixations
 - pupil dilation during long fixations
 - HR/HRV changes around fixation
-

Processing Requirements

- remove invalid gaze samples
- exclude EyesNotFound periods
- aggregate by Eye movement type index

Limitations

- fixation duration alone cannot distinguish visual vs cognitive difficulty
- influenced by text complexity
- influenced by fatigue and attention

Feature 2 - Saccade Duration

Classification: Primarily Visual / Oculomotor

Definition

Duration of a saccadic eye movement between fixation points.

Computed from:

Eye movement type = Saccade

Gaze event duration [ms]

Interpretation

Visual / Oculomotor Component

Abnormal saccade timing may indicate:

- unstable reading behavior
 - inefficient gaze transitions
 - disrupted visual tracking
 - increased effort navigating disappearing text
-

Cognitive Component

Saccades may also be indirectly influenced by:

- attentional shifts
- reading strategy
- processing difficulty
- auditory pacing adaptation

However, saccades are generally interpreted as more visually and oculomotor-driven than fixations.

Task-Level Analysis

mean_saccade_duration

saccade_time_percentage

saccade_duration_variability

Usage

Compare saccadic behavior across:

- amblyopia vs controls
 - left vs right amblyopia
 - task vs task
-

Event-Level Analysis

Definition

Specific saccades that significantly differ from the participant/task baseline.

Potential event-centered analysis:

- EEG activity surrounding long or unstable saccades
- pupil changes before/after saccades
- physiological responses during disrupted reading transitions
- regressions following abnormal saccades

Processing Requirements

- keep only Eye movement type = Saccade
- remove invalid segments
- aggregate using Eye movement type index
- normalize by participant baseline
- separate forward vs regression saccades

Limitations

- short noisy movements may be misclassified
- interpretation depends on reading context

Feature 3 - EyesNotFound Bursts

Classification: Primarily Visual / Oculomotor

Definition

Periods in which the Tobii system temporarily fails to detect the participant's eyes for a continuous duration.

Computed from:

Eye movement type = EyesNotFound

An EyesNotFound burst is defined as a continuous sequence of consecutive EyesNotFound events exceeding a predefined duration threshold.

Interpretation

Visual / Oculomotor Component

EyesNotFound bursts may reflect:

- unstable gaze behavior
 - excessive blinking
 - head movement
 - tracking instability
 - visual fatigue
 - difficulty maintaining stable fixation during reading
-

Possible Cognitive / Attentional Component

Longer or more frequent EyesNotFound bursts may also reflect:

- attentional disengagement
- task overload
- frustration
- temporary withdrawal from the reading task

Interpretation should be made cautiously, since tracking loss can also result from technical factors.

Task-Level Analysis

EyesNotFound_percentage

mean_EyesNotFound_duration

EyesNotFound_duration_variability

Usage

Compare tracking stability across:

- amblyopia vs controls
 - left vs right amblyopia
 - task vs task
-

Event-Level Analysis

Definition

Specific EyesNotFound bursts exceeding participant/task baseline duration.

Potential event-centered analysis:

- EEG activity before and after gaze loss
- pupil behavior preceding tracking loss
- physiological stress around disengagement events

- relationship between tracking loss and regressions
-

Processing Requirements

- identify consecutive EyesNotFound samples
- merge continuous segments into single events
- calculate event duration

Suggested threshold

Example:

EyesNotFound_duration > 100 ms

Threshold should later be validated empirically.

Limitations

- may reflect technical tracking failure rather than participant behavior
- highly sensitive to head movement and blinking
- affected by calibration quality and lighting conditions
- should not be interpreted as a direct cognitive measure alone

Feature 4- Pupil Diameter

Classification: Primarily Cognitive / Attentional

Secondary physiological and visual component.

Definition

Pupil diameter represents the measured size of the participant's pupils during the reading task.

Computed from:

Pupil diameter left [mm]

Pupil diameter right [mm]

Pupil size is treated as a dynamic physiological marker that may reflect cognitive load, attentional effort, arousal, and task engagement during reading.

Interpretation

Cognitive / Attentional Component

Increased pupil diameter may be associated with:

- increased cognitive load
 - attentional effort
 - processing difficulty
 - task engagement
 - increased reading demand
-

Visual Component

Pupil size may also be influenced by:

- visual strain
- unstable fixation
- prolonged visual processing
- gaze instability

Interpretation should consider both cognitive and visual factors.

Task-Level Analysis

mean_pupil_diameter

pupil_diameter_variability

Usage

Compare pupil behavior across:

- amblyopia vs controls
 - left vs right amblyopia
 - task vs task
-

Processing Requirements

- remove invalid samples
- exclude blink-related signal loss
- synchronize left/right pupil measurements
- remove extreme outliers
- smooth short fluctuations
- normalize per participant

Limitations

- highly sensitive to lighting conditions
- affected by blinks and tracking instability
- may fluctuate independently of cognitive load
- left/right asymmetry may reflect tracking artifacts rather than physiological differences

Feature 5 - Regression Events

Classification: Visual / Oculomotor + Cognitive

Definition

Regression events are backward gaze transitions occurring during reading.

Computed from:

Eye movement type

Eye movement type index

Fixation point X [DACS px]

Fixation point Y [DACS px]

Gaze event duration [ms]

Interpretation

Visual / Oculomotor Component

Regressions may reflect:

- unstable visual scanning
 - inefficient reading progression
 - gaze-control instability
 - loss of place during reading
 - difficulty processing disappearing text
-

Cognitive Component

Regressions may also reflect:

- increased processing difficulty
 - rereading behavior
 - attentional lapses
 - lexical or semantic uncertainty
 - increased cognitive load
-

Task-Level Analysis

regression_percentage

mean_regression_distance

regression_duration_variability

Regression percentage is calculated relative to the total number of saccadic transitions rather than total task duration, since regression events are brief.

Usage

Compare regression behavior across:

- amblyopia vs controls
 - left vs right amblyopia
 - task vs task
-

Event-Level Analysis

Definition

Each detected regression event is treated as a reading-disruption event.

Potential event-centered analysis:

- EEG activity surrounding regressions

Processing Requirements

Required preprocessing

- keep only valid fixation transitions
- reconstruct fixation order using Eye movement type index
- remove invalid gaze samples
- identify same-line transitions using fixation Y coordinates

Basic Detection Logic

For Hebrew reading:

$$\Delta x = fixation_{next} - fixation_{previous}$$

$\Delta x < 0 \rightarrow$ forward reading movement

$\Delta x > 0 \rightarrow$ regression

Regression classification should only be applied within the same text line.

Limitations

- regression detection depends on accurate fixation classification
- noisy gaze coordinates may produce false regressions
- line transitions may be incorrectly classified as regressions
- thresholds for large regressions require empirical validation

Feature 6- Validity Integration

Classification: Eye-Tracking Quality Control

Validity left

Validity right

Eye movement type

Purpose

Validity fields are treated as dynamic sample-level reliability indicators and are integrated into all eye-tracking analyses.

Unlike session-level calibration metrics, validity measures are evaluated continuously throughout the task and are therefore used as the primary quality-control mechanism for event-level and task-level analysis.

General Principle

All eye-tracking features should be interpreted together with:

- left-eye validity
- right-eye validity

The goal is to distinguish:

- true behavioral/cognitive effects
from:
 - unreliable gaze acquisition
 - blinking
 - head movement
-

Task-Level Quality Control

left_valid_percentage

right_valid_percentage

combined_valid_percentage

Usage

Task-level validity metrics are used to:

- estimate overall recording reliability
 - identify unstable sessions
 - detect excessive gaze loss during difficult tasks
-

Exclusion Criteria (to be validated empirically)

Examples:

combined_valid_percentage < threshold

Thresholds should be determined after examining the dataset distribution.

Event-Level Quality Control

Validity measures must also be evaluated around specific events such as:

- long fixations
- regressions
- pupil dilation events
- EyesNotFound bursts

Event-Level Validation Logic

Example:

```
if validity_left == invalid
or validity_right == invalid:
    exclude_or_flag_event
```

Integration With Other Features

Fixation Analysis

Long fixations should only be interpreted if gaze validity remains stable throughout the fixation event.

Regression Detection

Regression classification should only be computed from valid fixation transitions. Invalid gaze samples may produce false regressions.

Pupil Diameter

Pupil measurements should only be analyzed during periods of valid tracking. Blink-related invalid samples must be excluded before pupil analysis.

Limitations

- validity reflects tracking reliability, not visual ability
- invalid samples may result from technical factors
- left/right asymmetry may reflect calibration differences rather than physiological asymmetry
- excessive filtering may remove meaningful behavioral events

Validity measures should therefore be treated as contextual quality indicators rather than standalone behavioral outcomes.

Corsano Database

Export files from [Corsano research portal](#) in the following format:

Export Data

- | | |
|--|---|
| ☒ PR/BRPM/Activity | ☐ Workout |
| ☐ Respiration | ☐ Sleep |
| ☐ RR intervals | ☒ HRV |
| ☐ NIBP | ☐ Temperature |
| ☐ Emography | |

Raw Data

- | | |
|---|--|
| ☐ PPGG | ☐ PPG G/R/Ir |
| ☐ ECG | ☐ BioZ |
| ☒ ACC xyz | ☐ Raw temperature |

Corsano ACC Database

Source File

- acc.xlsx

Purpose of the Dataset

The ACC dataset contains accelerometer-based motion data collected from the Corsano wearable device during the experimental session. This dataset is primarily used for movement monitoring, artifact detection, and physiological data quality assessment during reading and resting-state tasks.

The accelerometer signals can help identify:

- participant movement
- restlessness
- body motion during reading
- motion-related physiological artifacts
- unstable recording segments

This dataset may also be used for synchronization and behavioral state analysis across experimental conditions.

Relevant Fields

timestamp

Unix timestamp in milliseconds.

Primary time reference used for synchronization and temporal alignment with other datasets.

date

Human-readable date and time value.

Contains recording date, time, and timezone information.

metric_id

Corsano internal metric identifier.

Used to identify the measurement type.

chunk_index

Sequential data chunk index.

Used internally for data segmentation and stream organization.

quality

Signal quality indicator.

Used for filtering low-quality or unreliable recording segments.

body_pose

Body posture / pose classification provided by Corsano.

May indicate participant posture or movement state during recording.

accX

Accelerometer X-axis value.

Represents motion along the X axis.

accY

Accelerometer Y-axis value.

Represents motion along the Y axis.

accZ

Accelerometer Z-axis value.

Represents motion along the Z axis.

Feature 7- ACC Motion Magnitude

Classification: Primarily Physiological / Movement Quality Control

Definition

Motion magnitude represents the overall intensity of participant movement measured by the accelerometer during the task.

Computed from:

accX

accY

accZ

Using:

$$\text{Motion Magnitude} = \sqrt{\text{accX}^2 + \text{accY}^2 + \text{accZ}^2}$$

This feature combines movement across all three accelerometer axes into a single motion-intensity measure.

Interpretation

Physiological / Recording Component

Increased motion magnitude may reflect:

- participant movement
 - restlessness
 - posture adjustment
-

Behavioral Component

Movement changes may also occur during:

- difficult reading conditions
- frustration
- attentional disengagement
- increased cognitive effort
- oral reading conditions

Interpretation should be made cautiously, since movement does not directly indicate cognitive processing.

Task-Level Analysis

mean_motion_magnitude

motion_variability

high_motion_percentage

motion_burst_count

Usage

Compare movement behavior across:

- amblyopia vs controls
- left vs right amblyopia
- task vs task

Motion features may also be used to estimate:

- task-related restlessness
- recording stability
- movement-related physiological contamination

Processing Requirements

Required preprocessing

- synchronize accelerometer timestamps with task periods
- remove corrupted or missing samples
- calculate motion magnitude

Suggested processing

- smooth short motion spikes
- normalize per participant
- define movement thresholds empirically

Example:

high_motion_event = motion_magnitude > participant_baseline

Quality Control Integration

Motion magnitude should be evaluated together with:

bpm_q

resp_q

Validity left/right

The goal is to distinguish:

- true physiological/cognitive effects from movement-related artifacts
-

Limitations

- movement does not directly reflect cognitive state
- accelerometer sensitivity depends on wearable placement
- small movements may produce large accelerometer changes
- motion artifacts may overlap with genuine behavioral responses

Corsano Activity Database

Source File

- activity.xlsx

Purpose of the Dataset

The Activity dataset contains physiological and respiratory measurements collected from the Corsano wearable device during the experimental session.

This dataset is primarily used for:

- heart-rate monitoring during reading tasks
 - physiological state analysis
 - respiration analysis
 - task engagement analysis
 - synchronization with EEG, eye-tracking, and behavioral datasets
-

Relevant Fields

timestamp

Unix timestamp in milliseconds.

Primary temporal reference used for synchronization across datasets.

date

Human-readable date and time value.

Contains recording date, time, and timezone information.

bpm

Heart rate value in beats per minute.

Primary physiological measure used during the experiment.

respiration_rate

Estimated breathing rate.

Used for physiological and arousal-related analysis.

bpm_q

Heart-rate quality score provided by Corsano.

Quality values range from:

- 0 = very low quality
- 1 = low quality
- 2 = fair quality
- 3 = good quality
- 4 = very good quality

The score reflects signal reliability and measurement confidence based on the underlying PPG signal quality and motion stability.

resp_q

Respiration-rate quality score provided by Corsano.

Quality values range from:

- 0 = very low quality
- 1 = low quality
- 2 = fair quality
- 3 = good quality

- 4 = very good quality

The score reflects the reliability of the respiration estimation generated from the physiological signal.

ACC signals are used as contextual motion indicators and artifact-reference signals, while bpm_q and resp_q are treated as device-generated physiological signal confidence measures. Combined analysis of motion and quality metrics enables identification of unstable physiological recording segments during experimental tasks.

Feature 8- Heart Rate (BPM)

Classification: Primarily Physiological

Secondary cognitive and attentional component.

Definition

Heart rate represents the number of heartbeats per minute recorded during the experimental task.

Computed from:

bpm

Heart rate is treated as a dynamic physiological marker reflecting autonomic activation during reading and resting-state conditions.

Interpretation

Physiological Component

Changes in heart rate may reflect:

- autonomic activation
 - physiological arousal
 - stress response
 - task intensity
-

Cognitive / Attentional Component

Heart rate changes may also occur during:

- increased cognitive effort
- difficult reading conditions
- attentional engagement
- oral reading
- audiovisual pacing conditions

Interpretation should be made cautiously, since heart rate is influenced by multiple physiological factors.

Task-Level Analysis

mean_bpm

bpm_change_from_baseline

Usage

Compare physiological activation across:

- amblyopia vs controls
- left vs right amblyopia
- task vs silent task

Heart-rate changes may help identify:

- increased reading effort
 - task-related arousal
 - stress-related physiological responses
-

Processing Requirements

- synchronize timestamps with task periods
- remove low-quality samples using bpm_q
- smooth short-term fluctuations

- normalize per participant
- compute baseline-relative changes

Example:

bpm_change_from_baseline = current_bpm - resting_baseline_bpm

Quality Control Integration

Heart-rate analysis should always be interpreted together with:

bpm_q

ACC motion magnitude

Low-quality physiological segments or excessive movement may produce unreliable BPM measurements.

Limitations

- heart rate is not specific to cognitive load
- influenced by movement, respiration, stress, and emotional state
- short-term fluctuations may not reflect meaningful task-related changes
- oral reading may increase BPM independently of reading difficulty

Feature 9- Respiration Rate

Classification: Physiological + Cognitive / Attentional

Definition

Respiration rate represents the estimated breathing frequency during the experimental task.

Computed from:

respiration_rate

Respiration is treated as a physiological rhythm potentially associated with arousal, attentional state, and auditory pacing.

Interpretation

Physiological Component

Changes in respiration rate may reflect:

- autonomic activation
 - physiological arousal
 - stress response
 - task-related regulation
-

Cognitive / Attentional Component

Respiration may also change during:

- increased attentional demand
 - difficult reading conditions
 - metronome synchronization
 - cognitive effort
 - oral reading
-

Task-Level Analysis

mean_respiration_rate

respiration_variability

respiration_change_from_baseline

Usage

Compare respiratory behavior across:

- amblyopia vs controls
- left vs right amblyopia
- task vs task

Potential goals:

- identify arousal-related changes
 - examine auditory pacing effects
 - investigate physiological regulation during reading
-

Processing Requirements

- synchronize respiration timestamps with task periods
- remove unreliable samples using resp_q
- smooth short fluctuations
- normalize relative to resting baseline

Quality Control Integration

Respiration analysis should always be interpreted together with:

resp_q

ACC motion magnitude

Movement artifacts and unstable physiological recordings may affect respiration estimates.

Limitations

- breathing changes may result from speech, movement, or stress
- oral reading may strongly affect respiration patterns
- short-term respiration changes may be noisy

Corsano HRV Database

Source File

- heart_rate_variability.xlsx

Purpose of the Dataset

The HRV dataset contains heart-rate variability metrics exported from the Corsano platform. It is used to evaluate autonomic regulation, physiological stability, and possible task-related changes during the experiment.

Relevant Fields

timestamp

Unix timestamp in milliseconds.

Primary temporal reference for synchronization.

date

Human-readable date and time.

beats

Number of detected heartbeats in the analysis window.

mrr

Mean RR interval.

sdnn

Standard deviation of RR intervals.

rmssd

Root mean square of successive RR interval differences.

pnn50

Percentage of successive RR intervals differing by more than 50 ms.

hf

High-frequency HRV component.

lf

Low-frequency HRV component.

vlf

Very-low-frequency HRV component.

hrv

Corsano-derived general HRV value.

si

Stress index.

sin

Normalized stress index / stress-related value.

sic

Color/status code associated with the stress index.

Feature 10- RMSSD

Classification: Primarily Physiological + Cognitive / Attentional

Definition

RMSSD (Root Mean Square of Successive Differences) is a heart-rate variability measure reflecting short-term variability between consecutive heartbeats.

Computed from:

rmssd

RMSSD is treated as a marker of autonomic regulation and short-term physiological adaptability during reading tasks.

Interpretation

Physiological Component

RMSSD may reflect:

- autonomic flexibility
 - physiological regulation
 - parasympathetic activity
 - adaptive physiological state
-

Cognitive / Attentional Component

Reduced RMSSD may also occur during:

- increased cognitive load
 - attentional demand
 - stress
 - difficult reading conditions
 - disrupted reading flow
-

Task-Level Analysis

mean_rmssd

rmssd_variability

rmssd_change_from_baseline

Usage

Compare autonomic regulation across:

- amblyopia vs controls
- left vs right amblyopia
- task vs task

Potential goals:

- identify increased physiological effort
 - estimate cognitive load during reading
 - examine regulation differences between groups
-

Processing Requirements

- synchronize HRV windows with task periods
- remove low-quality physiological segments
- normalize relative to resting baseline

Example:

`rmssd_change_from_baseline = current_rmssd - resting_rmssd`

Quality Control Integration

RMSSD analysis should always be interpreted together with:

bpm_q

ACC motion magnitude

Movement artifacts and unstable PPG recordings may affect HRV estimates.

Limitations

- influenced by respiration and movement
 - interpretation should be cautious during oral reading conditions
-

Feature 11- Stress Index (SI)

Classification: Physiological + Cognitive / Attentional

Definition

Stress Index (SI) is a Corsano-derived physiological metric intended to estimate autonomic stress and physiological activation during the experimental task.

Computed from:

si

Interpretation

Physiological Component

Increased SI may reflect:

- physiological stress
 - autonomic activation
 - increased arousal
-

Cognitive / Attentional Component

Elevated SI may also occur during:

- cognitively demanding reading
 - audiovisual pacing conditions
 - disrupted reading flow
 - attentional overload
 - difficult visual processing
-

Task-Level Analysis

mean_si

si_variability

si_change_from_baseline

Usage

Compare physiological stress across:

- amblyopia vs controls
- left vs right amblyopia
- task vs task

Potential goals:

- identify stressful reading conditions
 - estimate autonomic load during reading
 - examine regulation differences between groups
-

Processing Requirements

- synchronize SI timestamps with task periods
- remove low-quality physiological segments
- normalize relative to resting baseline
- smooth short fluctuations

Quality Control Integration

SI analysis should always be interpreted together with:

bpm_q

resp_q

ACC motion magnitude

Limitations

- influenced by movement and physiological noise
- should not be interpreted as a clinical stress measure

EEG Analysis Framework

EEG Analysis Structure

The EEG component of the project is analyzed at two complementary levels:

1. Task-Level EEG Analysis

Task-level analysis examines neural activity across entire experimental conditions. These features are used to characterize overall neural states during different reading paradigms and to compare groups and task conditions.

2. Event-Level EEG Analysis

Event-level analysis examines neural activity surrounding behaviorally defined reading events extracted from the eye-tracking system, such as:

long_fixation_events

regression_events

EyesNotFound_bursts

Rather than focusing on predefined ERP components, the project aims to identify neural patterns associated with disrupted reading behavior, cognitive effort, gaze instability, and rereading events.

EEG Preprocessing Pipeline

Because EEG recordings are initially acquired as raw signals, preprocessing is required before feature extraction and analysis.

Planned preprocessing stages

raw EEG import

channel selection

band-pass filtering

bad channel detection

artifact removal

movement/blink artifact handling

task segmentation

event-centered epoch extraction

baseline normalization

frequency analysis

connectivity analysis

Eye-tracking and physiological events may later be aligned with EEG segments for multimodal event-centered analysis.

Task-Level EEG Analysis

Feature 12 - Frontal Theta Power

Classification: Primarily Cognitive / Attentional

Definition

Frontal theta power represents oscillatory activity within the theta frequency range measured over frontal EEG regions.

Typical theta range: 4–8 Hz

Interpretation

Frontal theta activity is commonly associated with:

- cognitive effort
- attentional control
- executive processing
- working memory load
- task engagement
- conflict monitoring

In the context of reading tasks, increased frontal theta power may reflect:

- increased cognitive demand
 - effortful reading
 - attentional recruitment
 - adaptation to difficult task conditions
-

Task-Level Analysis

mean_frontal_theta_power

theta_power_change_from_baseline

theta_power_variability

Usage

Compare frontal theta activity across:

- amblyopia vs controls
- left vs right amblyopia
- task vs static task

Potential goals:

- estimate cognitive effort during reading
 - identify attentional recruitment
 - characterize task difficulty
-

Processing Requirements

- filtering within theta band
 - artifact removal
 - frontal channel selection
 - power estimation within defined task windows
-

Limitations

- influenced by fatigue and general attention
- sensitive to movement and blink artifact

Feature 13 - Occipital Alpha Power

Classification: Primarily Visual / Attentional

Definition

Occipital alpha power represents oscillatory activity within the alpha frequency range measured over occipital visual regions.

Typical alpha range: 8–12 Hz

Interpretation

Occipital alpha activity is commonly associated with:

- visual processing state
- attentional allocation
- visual engagement
- sensory inhibition

Reduced occipital alpha power may reflect:

- increased visual processing
- increased visual attention
- greater visual task engagement

Increased alpha power may reflect:

- reduced visual engagement
 - visual disengagement
-

Task-Level Analysis

Possible Features

mean_occipital_alpha_power

alpha_power_change_from_baseline

alpha_power_variability

Usage

Compare occipital alpha activity across:

- amblyopia vs controls
- left vs right amblyopia
- task vs task

Potential goals:

- characterize visual engagement during reading
 - examine visual processing demands
 - investigate visual-attentional differences between groups
-

Processing Requirements

- alpha-band filtering
 - occipital channel selection
 - artifact removal
 - power estimation within task windows
-

Limitations

- alpha activity is sensitive to eye closure and blinking
- influenced by general vigilance and fatigue
- visual interpretation depends on eye-tracking validity

Feature 14 - Frontal Theta / Occipital Alpha Ratio

Classification: Cognitive / Attentional + Visual Integration

Definition

The frontal theta / occipital alpha ratio represents the relationship between frontal cognitive-control activity and occipital visual-processing activity during reading tasks.

Computed from:

frontal_theta_power

occipital_alpha_power

Formula:

$$\text{Theta/Alpha Ratio} = \frac{\text{Frontal Theta Power}}{\text{Occipital Alpha Power}}$$

Interpretation

A higher frontal theta / occipital alpha ratio may reflect:

- increased cognitive effort
- increased attentional control
- higher mental workload
- stronger executive recruitment relative to visual-state regulation

In reading tasks, this ratio may indicate:

- effortful reading
 - increased cognitive compensation
 - imbalance between visual processing and cognitive control systems
-

Task-Level Analysis

Possible Features

mean_theta_alpha_ratio

theta_alpha_ratio_change_from_baseline

theta_alpha_ratio_variability

Usage

Compare ratio values across:

- amblyopia vs controls
- left vs right amblyopia
- task vs task

Potential goals:

- identify cognitively demanding reading conditions
 - examine compensatory cognitive recruitment
 - characterize visual-cognitive interaction during reading
-

Processing Requirements

- compute frontal theta power
- compute occipital alpha power
- calculate ratio per task window
- normalize relative to resting-state baseline if possible
- remove blink and movement artifacts before spectral analysis

Limitations

- ratio values depend on preprocessing quality
- alpha power is sensitive to eye closure and blinking
- not specific to reading alone
- influenced by fatigue, vigilance, and attentional state

Feature 15 - Theta PLV: Occipital ↔ Temporal

Classification: Audiovisual Integration

Definition

Theta-band Phase Locking Value (PLV) between occipital and temporal EEG regions.

Computed from:

theta-band EEG activity

occipital electrodes

temporal electrodes

Typical theta range: 4–8 Hz

PLV measures the consistency of phase synchronization between neural signals recorded from occipital and temporal regions during reading tasks.

Interpretation

Occipital–temporal theta PLV may reflect:

- audiovisual coordination
- phonological processing integration
- reading-network synchronization

Higher PLV may indicate stronger synchronization between visual and temporal processing networks during reading.

Task-Level Analysis

mean_occipital_temporal_theta_plv

OT_plv_change_from_baseline

OT_plv_variability

Usage

Compare occipital–temporal synchronization across:

- amblyopia vs controls
- left vs right amblyopia
- task vs task

Potential goals:

- examine audiovisual integration during reading
 - investigate visual-language coordination
 - identify disrupted reading-network synchronization
-

Processing Requirements

- theta-band filtering
 - occipital and temporal channel selection
 - artifact removal
 - phase extraction
 - PLV computation within task windows
-

Limitations

- PLV is sensitive to noise and volume conduction
- does not indicate directionality of information flow
- interpretation depends on channel selection

Feature 16 - Theta PLV: Temporal ↔ Frontal

Classification: Cognitive / Attentional + Language Integration

Definition

Theta-band Phase Locking Value (PLV) between temporal and frontal EEG regions.

Computed from:

theta-band EEG activity

temporal electrodes

frontal electrodes

Interpretation

Temporal–frontal theta PLV may reflect:

- cognitive control during reading
- attentional coordination
- executive-language interaction
- reading effort
- working-memory involvement during reading

Higher PLV may indicate increased coordination between language-processing and executive-control systems.

Task-Level Analysis

mean_temporal_frontal_theta_plv

TF_plv_change_from_baseline

TF_plv_variability

Usage

Compare temporal–frontal synchronization across:

- amblyopia vs controls
- left vs right amblyopia
- task vs task

Potential goals:

- examine cognitive recruitment during reading
 - investigate executive involvement during difficult reading conditions
 - characterize attentional coordination
-

Processing Requirements

- theta-band filtering
 - frontal and temporal channel selection
 - artifact removal
 - phase extraction
 - PLV computation within task windows
-

Limitations

- may be affected by movement and blink artifacts
- does not directly indicate causal interaction
- influenced by general attentional state

Feature 17 - Theta PLV: Occipital ↔ Frontal

Classification: Visual + Cognitive / Attentional Integration

Definition

Theta-band Phase Locking Value (PLV) between occipital and frontal EEG regions.

Computed from:

theta-band EEG activity

occipital electrodes

frontal electrodes

Interpretation

Occipital–frontal theta PLV may reflect:

- top-down attentional regulation
- visual-attentional coordination
- visual cognitive-control interaction
- effortful visual processing
- attentional guidance during reading

Higher PLV may indicate increased coordination between visual and executive systems during difficult reading conditions.

Task-Level Analysis

Possible Features

mean_occipital_frontal_theta_plv

OF_plv_change_from_baseline

OF_plv_variability

Usage

Compare occipital–frontal synchronization across:

- amblyopia vs controls
- left vs right amblyopia
- task vs task

Potential goals:

- examine attentional regulation during reading
 - investigate visual-cognitive interaction
 - characterize compensatory recruitment during difficult reading
-

Processing Requirements

- theta-band filtering
 - occipital and frontal channel selection
 - artifact removal
 - phase extraction
 - PLV computation within task windows
-

Limitations

- influenced by eye-movement artifacts
- synchronization does not necessarily imply functional causation

Event-Level EEG Analysis

General Approach

Event-level EEG analysis is based on behaviorally defined events extracted from the eye-tracking dataset rather than predefined stimulus-locked ERP components.

The goal of event-level EEG analysis is to determine whether specific reading events are consistently associated with characteristic neural activity patterns across trials and participants.

Core eye-tracking events used for EEG-centered analysis:

long_fixation_events

regression_events

EyesNotFound_bursts

Event Definition

Each event is assigned an:

event_onset_time

The onset time is treated as: $t = 0$ ms
for EEG epoch extraction.

The beginning of the event is used as the temporal anchor because:

- event durations vary substantially
- some events are very short
- fixed-duration EEG windows provide more stable spectral estimation and cross-event comparison

The full event duration may later be used as an additional contextual feature, but not as the primary EEG analysis window.

EEG Epoch Extraction

For each detected event, an EEG segment is extracted around the event onset.

Planned EEG windows

pre-event window:

-500 ms to 0 ms

event window:

0 ms to event_duration

post-event window:

event_end to +500 ms

The event duration itself is included within the analyzed window.

The complete analysis window therefore includes:

- pre-event activity
 - activity during the event
 - immediate post-event activity
-

EEG Features Extracted Around Events

The same EEG features used for task-level analysis are extracted within each event-centered EEG window.

Extracted Features

frontal_theta_power
occipital_alpha_power
frontal_theta_occipital_alpha_ratio
theta_PLV_occipital_temporal
theta_PLV_temporal_frontal
theta_PLV_occipital_frontal

Event-Level Processing

For each individual event:

1. Extract EEG data from the event-centered time window.
 2. Compute all predefined EEG features within the window.
 3. Store feature values together with:
 - event type
 - task
 - participant
 - event timing information
-

Event-Level Aggregation

Events are grouped by event category:

long_fixation_events
regression_events
EyesNotFound_bursts

For each event category, the system computes:

number_of_events
mean_event_duration
event_duration_variability
mean_EEG_features
feature_standard_deviation
feature_variance

Baseline Comparison Strategy

Event-level EEG features are compared against the participant's mean EEG activity during the same task condition.

Example:

event_frontal_theta_power
vs

mean_task_frontal_theta_power

This comparison is used to determine whether specific event categories are consistently associated with characteristic neural patterns relative to the participant's ongoing reading state.

Processing Requirements

Required preprocessing:

artifact removal
blink correction
movement artifact handling
event timestamp synchronization
task segmentation
EEG epoch extraction
frequency filtering
spectral estimation
PLV computation

Limitations

- event durations vary substantially
 - eye movements generate EEG artifacts
 - synchronization precision is critical
 - EEG estimation may become unstable for extremely short events
 - event-level analysis is exploratory and highly preprocessing-dependent
-

Purpose of Each Feature in Event-Level Analysis

Frontal Theta Power

Used to estimate:

- cognitive effort
- attentional recruitment
- executive-control activation

Potential question:

Does frontal theta increase around regressions or prolonged fixations?

Occipital Alpha Power

Used to estimate:

- visual-processing state
- visual attention
- visual engagement

Potential question:

Does visual-processing demand change during difficult reading events?

Frontal Theta / Occipital Alpha Ratio

Used to estimate:

- cognitive-control demand relative to visual-processing activity

Potential question:

Do difficult reading events require increased cognitive effort relative to visual processing?

Theta PLV — Occipital ↔ Temporal

Used to estimate:

- visual-language synchronization
- audiovisual integration

Potential question:

Do regressions occur during reduced visual-language coordination?

Theta PLV — Temporal ↔ Frontal

Used to estimate:

- executive-language coordination
- attentional recruitment during reading

Potential question:

Do difficult reading events involve increased frontal-language synchronization?

Theta PLV — Occipital ↔ Frontal

Used to estimate:

- visual-attentional coordination
- top-down attentional regulation

Potential question:

Are disrupted reading events associated with altered visual-attentional synchronization?

Code Implementation Notes - Processing Status

The system should maintain an internal processing-status layer for each participant and each uploaded task.

This status layer is not displayed as a main interface component, but is used internally to determine whether each analysis can be generated or loaded from existing saved results.

For each task, the system should internally track whether the required files exist and whether each analysis table has already been generated.

Example internal status fields:

eye_tracking_uploaded

acc_uploaded

activity_uploaded

hrv_uploaded

eeg_uploaded

task_level_features_generated

event_level_features_generated

quality_control_generated

Before running any analysis, the system should check whether all required input files for that analysis are available.

If required files are missing, the system should not crash. Instead, it should display a clear error message specifying which file or dependency is missing.

Examples:

- "EEG analysis cannot be generated because Task.eeg is missing."
- "Eye-tracking analysis cannot be generated because EyeTracking.xlsx is missing."
- "Baseline comparison cannot be generated because Resting state data is missing."

Missing files should only block the specific analysis that depends on them, not the entire participant record.

The system must support modular analysis. Each participant may contain any subset of available modalities, including EEG-only, eye-tracking-only, Corsano-only, or multimodal recordings.

Analysis modules should run only when their required input files are available. Missing modalities should not prevent the user from analyzing existing data from other modalities.

Participant and Task Storage Structure

When a new participant is created, the system automatically creates a participant folder named by the participant ID.

Participant-level metadata should not be stored inside the participant folder as the primary record. Instead, the following general participant details should be stored in a centralized participants table:

participant_id

participant_name

participant_age

participant_group

notes

Each time a new participant is added, a new row is appended to this centralized table.

Inside each participant folder, the system stores the uploaded recording files. Task-specific EEG files should be stored inside a separate task folder named according to the task name.

Example structure:

```
database/
├── participants_table.xlsx
├── participants/
│   └── participant_123456789/
│       ├── EyeTracking.xlsx
│       ├── acc.xlsx
│       ├── activity.xlsx
│       ├── heart_rate_variability.xlsx
│       └── Oral Reading - Erased Text/
│           ├── Task.ahdr
│           ├── Task.eeg
│           ├── Task.amrk
│           ├── reading_comprehension_score.txt
│           ├── table_3_EEG_Data.xlsx
│           └── table_5_Long_Fixation_Events.xlsx
```

Any generated analysis table that is related to a single task should be saved inside that task's folder.

All uploaded raw files must remain unchanged after upload.

The system should never overwrite raw recording files.

All preprocessing outputs, generated feature tables, quality-control tables, event-detection outputs, and derived analysis files should be saved as separate processed files inside the corresponding participant/task folders.

Synchronization process

E-Prime Task Synchronization Specification

Purpose:

The system must extract a continuous task time window from the E-Prime log file in order to synchronize eye-tracking, EEG, and physiological recordings.

REQUIRED FIELDS TO EXTRACT

The synchronization parser must extract the following fields from the uploaded E-Prime .txt file:

SessionStartDateTimeUtc

FixationStart.OnsetTime

FixationStart.OnsetToOnsetTime

SixationOpenS.OnsetToOnsetTime

The system must not assume fixed E-Prime file names.

The uploaded .txt file provided by the user should be treated as the E-Prime synchronization source for the selected task.

The task window is defined as:

START = appearance of the fixation cross ("+")

END = appearance of the smiley screen indicating task completion

The system should treat the entire reading task as one continuous stimulus window.

E-Prime Task Window Definition

The task window is defined using the following fields from the uploaded E-Prime .txt log file.

RELEVANT FIELDS

1. SessionStartDateTimeUtc

Example: SessionStartDateTimeUtc: 14/04/2026 8:56:27

Meaning: Absolute UTC time at which the E-Prime session started running on the computer.

2. FixationStart.OnsetTime

Example: FixationStart.OnsetTime: 33615

Meaning: Timestamp (in milliseconds from session start) corresponding to the appearance of the fixation-cross ("+") screen.

זהו זמן תחילת מסך הפלוס השחור שמופיע לפני תחילת הסיפור

This field defines:

TASK_START_MS

3. FixationStart.OnsetToOnsetTime

Example: FixationStart.OnsetToOnsetTime: 3566

Meaning: Time duration (milliseconds) from fixation-cross onset until the next screen begins.

זהו משך הזמן שבו מסך הפלוס מוצג עד שהסיפור מתחיל.

4. SixationOpenS.OnsetToOnsetTime

Example: SixationOpenS.OnsetToOnsetTime: 94429

Meaning: Time duration (milliseconds) from story onset until the next screen begins.

זהו משך הזמן שבו הסיפור מוצג על המסך עד למסך הסיום (הסמיילי).

TASK WINDOW CALCULATION

The beginning of the task window is defined as:

$$\text{TASK_START_MS} = \text{FixationStart.OnsetTime}$$

The task duration is defined as:

$$\text{TASK_DURATION_MS} = \text{FixationStart.OnsetToOnsetTime} + \text{SixationOpenS.OnsetToOnsetTime}$$

The end of the task window is calculated as:

$$\text{TASK_END_MS} = \text{TASK_START_MS} + \text{TASK_DURATION_MS}$$

TASK WINDOW INTERPRETATION

The task window represents:

START:
appearance of the fixation-cross ("+") screen

END:
appearance of the final smiley/end screen

The entire reading task is treated as one continuous stimulus window.
Internal story segments are not analyzed separately.

EEG Task Synchronization Specification

SOURCE FILES

The EEG recording is defined using three BrainVision files uploaded by the user:

1. .ahdr
2. .amrk
3. .eeg

The system must treat these three files as one EEG recording unit.

RELEVANT FILE: .amrk

The EEG recording start timestamp is extracted from the first marker entry in the .amrk file.

Relevant field example:

Mk1=New Segment,,1,1,0,20260414115951981525

Meaning: This marker defines the recording start timestamp of the EEG acquisition computer.

Timestamp format:

YYYYMMDDHHMMSSffffff

Example interpretation: 2026-04-14 11:59:51.981525

This value defines:

EEG_RECORDING_START_LOCAL_TIME

UTC CONVERSION RULE

The BrainVision recording timestamp uses the local system clock of the EEG acquisition computer.

All recordings in this project are collected in Israel.

During May recordings, Israel uses daylight saving time: UTC+3

Therefore, the synchronization layer must always convert EEG timestamps to UTC using:

EEG_RECORDING_START_UTC =
EEG_RECORDING_START_LOCAL_TIME - 3 hours

Example:

local EEG recording time:

2026-04-14 11:59:51

converted UTC time:

2026-04-14 08:59:51

All synchronization operations across systems must use UTC timestamps internally.

RELEVANT FILE: .ahdr

Relevant field:

SamplingInterval=2000

Meaning:

Sampling interval in microseconds.

The system should calculate EEG sampling rate as:

EEG_SAMPLING_RATE_HZ = 1,000,000 / SamplingInterval

Example:

$$1,000,000 / 2000 = 500 \text{ Hz}$$

SYNCHRONIZATION PURPOSE

The EEG recording start timestamp will later be aligned with:

1. E-Prime task timestamps
2. Eye-tracking timestamps
3. Physiological recordings

This synchronization enables extraction of EEG task windows corresponding to behavioral task periods.

IMPORTANT IMPLEMENTATION RULES

The system must dynamically parse uploaded BrainVision files.

The original raw EEG files must never be modified.

If one of the required BrainVision files is missing (.ahdr / .amrk / .eeg), the system must return a clear error message.

The system must treat uploaded EEG recordings as continuous recordings that are segmented later using synchronization timestamps.

Eye-Tracking Synchronization Specification

SOURCE FILE

The eye-tracking recording is uploaded as a Tobii Pro Lab Excel export file.

Required file format:

.xlsx

Multiple Eye-Tracking Files Per Participant

A participant may have more than one Tobii Pro Lab Excel export file.

Each eye-tracking file may contain a continuous recording session that includes one task, two tasks, or several tasks.

Therefore, for each E-Prime task window, the system must search across all uploaded eye-tracking files for that participant and identify which Tobii file contains the relevant task time window.

A Tobii file should be considered matching if the task window satisfies:

$\text{TASK_START_UTC} \geq \text{EYETRACKING_RECORDING_START_UTC}$

and

$\text{TASK_END_UTC} \leq \text{EYETRACKING_RECORDING_END_UTC}$

If the task window is found inside a Tobii file, the system should segment only the rows from that file that fall within the task window.

If no uploaded Tobii file contains the task window, the system must return a clear error message:

"No eye-tracking recording contains the selected task time window."

The system should use millisecond-resolution exports.

Microsecond-resolution exports are not required because eye-movement events occur on a slower timescale than EEG activity and do not require microsecond-level synchronization precision.

RELEVANT FIELDS

1. Recording start time UTC

Example: 09:00:32.220

Meaning: Absolute UTC timestamp corresponding to the beginning of the eye-tracking recording.

This field defines:

EYETRACKING_RECORDING_START_UTC

2. Recording timestamp [ms]

Example:

0

6

15

Meaning:

Relative timestamp in milliseconds from the beginning of the eye-tracking recording.

0 ms corresponds to the start of the Tobii recording.

This field defines the main eye-tracking timeline used for synchronization and task segmentation.

SYNCHRONIZATION STRATEGY

The eye-tracking system already provides UTC-based timestamps.

Therefore:

1. E-Prime timestamps
 2. Tobii timestamps
 3. EEG timestamps converted to UTC
- can all be aligned into one unified synchronization timeline.

IMPLEMENTATION REQUIREMENTS

1. All synchronization operations should use UTC timestamps internally.
2. The system must preserve millisecond-level timing precision.
3. The original Tobii export file must never be modified.
4. The eye-tracking recording should be treated as one continuous session recording that is segmented later using task windows extracted from E-Prime.

Corsano Physiological Synchronization Specification

SOURCE FILES

1. activity.xlsx

Used for heart rate and respiration.

Relevant synchronization field:

timestamp

Meaning: Unix timestamp in milliseconds. This is the primary time reference for synchronization.

Relevant analysis fields:

bpm

respiration_rate

bpm_q

resp_q

2. heart_rate_variability.xlsx

Used for HRV measures.

Relevant synchronization field:

timestamp

Meaning: Unix timestamp in milliseconds.

Relevant analysis fields:

rmssd

si

3. acc.xlsx

Used for motion / artifact control.

Relevant synchronization field:

timestamp

Meaning: Unix timestamp in milliseconds.

Relevant analysis fields:

accX

accY

accZ

quality

body_pose

UTC HANDLING

Corsano timestamp is a Unix timestamp in milliseconds.

Unix timestamps are UTC-based by definition.

Therefore:

`CORSANO_TIME_UTC` = timestamp converted from Unix milliseconds to UTC datetime

TASK SEGMENTATION

The system should segment Corsano data using the task window extracted from E-Prime:

`TASK_START_UTC`

`TASK_END_UTC`

For each Corsano file, keep only rows where:

`timestamp_utc >= TASK_START_UTC`

and

`timestamp_utc <= TASK_END_UTC`

IMPLEMENTATION REQUIREMENTS

1. Use timestamp as the primary synchronization field.
2. Do not use the human-readable date field for synchronization.
3. Use date only for debugging or human inspection.
4. Preserve the original uploaded Corsano files unchanged.
5. If timestamp is missing, the relevant Corsano analysis should fail with a clear error message.

Hypothesis Framework

H1 - Amblyopia vs. Typical Development

Children with amblyopia will show different reading patterns compared to typically developing children, reflected in behavioral, eye-tracking, physiological, and EEG measures.

H2 - Left vs. Right Amblyopia

Children with left-eye and right-eye amblyopia will show different reading-related profiles, potentially due to lateralized visual and reading networks.

H3 - Visual vs. Cognitive Profile

Reading difficulty in amblyopia may reflect different underlying profiles:

Visual / oculomotor difficulty

- more unstable fixations
- longer fixation duration
- more saccades/regressions
- more EyesNotFound / validity issues
- stronger difficulty when letters disappear

Cognitive / attentional difficulty

- comprehension decreases despite relatively stable gaze
- physiological stress/HRV changes
- EEG changes in attention/executive networks

H4 - Integration Profile

Some children may show a combined difficulty, where visual instability, physiological load, and neural connectivity changes appear together during demanding reading conditions.